

LineSight

Cold-Start Visual Inspection for Textile Rolls:
Anomaly Detection, Conformal Calibration, and ASTM D5430 Scoring

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Abstract. A textile mill onboards a new fabric construction every week. Supervised defect segmentation needs a labelled defect corpus for each construction, and that corpus does not exist at the moment it is needed. LineSight inverts the question: instead of learning what a defect looks like, it learns what *this* fabric’s normal looks like from roughly thirty defect-free images, in about two seconds of fitting and with no training loop, then flags everything unlike it — including defect types nobody has labelled. Detections are converted to millimetres through a printed fiducial tape, tracked across frames into single physical defects, and scored under ASTM D5430 into a points total and a pass/hold/reject verdict. The detection threshold is not chosen by hand: the operator states a false-alarm budget and the threshold is the corresponding split-conformal quantile of scores on held-out clean fabric. On MVTec AD the shipped cold-start configuration reaches 0.962 image AUROC on *carpet* and 0.984 on *leather* against a published ~ 0.98 and ~ 1.00 obtained with a four-times larger backbone. On real woven fabric the system reaches 0.971 pixel AUROC and 0.859 tile AUROC across six AITEX constructions, against 0.629 when a bank is borrowed from a different construction — a gap that is the measured justification for the per-SKU design. The system is validated across four independent datasets, and its two characterised detection boundaries — sparse regular structure at the shipped input resolution, and calibration drawn from a different physical sample than the one being inspected — are diagnosed, bounded, and handled by the shipped configuration and the operating procedure.

1. Vision and Problem Statement

1.1 The problem, stated precisely

Woven and knitted fabric leaves a loom with defects: broken picks, missing ends, slubs, holes, oil stains, reed marks, float threads. Before a roll is cut into garments it is inspected, graded and either accepted, held or rejected. The grading is not subjective folklore; it is a standard. Under ASTM D5430 [1], every defect is scored by its *length* along the roll — one point up to 75 mm, two to 150, three to 230, four beyond — capped at four points per linear metre, and normalised to points per 100 square yards. A buyer contract then sets the threshold at which a roll is held or rejected.

Two facts about that process define this project. First, **the verdict depends on a physical measurement, not on a defect's name**. A mill does not need to know that a flaw is a slub rather than a float; it needs to know it is 180 mm long, 1.8 m down the roll, and therefore worth three points. Second, **the inspection is done by people**, at a light-box, for a full shift, and their performance falls with fatigue and varies between individuals. The measurement that decides commercial acceptance is taken by the least repeatable instrument in the mill.

1.2 Why the obvious automation does not work

The published literature on automated fabric inspection is overwhelmingly supervised: a convolutional network is trained on images of defects, labelled by class or segmented pixel by pixel. This works, and reported accuracies are high. It also fails at exactly the moment a mill needs it.

A mill running buyer-specific orders introduces new fabric constructions continuously — different yarn counts, weaves, finishes, colours. Each construction changes what “normal” looks like to a camera, and therefore what a defect looks like against it. A supervised model needs a labelled corpus *per construction*, collected by photographing and annotating defects that have not happened yet in a fabric the mill has not yet run. That corpus does not exist at onboarding, and waiting to accumulate one means the system is useless precisely during the production run it was bought to protect. This is the **cold-start problem**, and it is the problem LineSight exists to solve.

1.3 The inversion

Instead of asking *what does a defect look like*, LineSight asks *what does this SKU's normal fabric look like*. Defect-free fabric is the one thing every mill has in abundance: thirty photographs of good cloth can be taken in a minute by a machine operator with no annotation, no labelling tool and no expertise. From those, the system builds a compact statistical profile of normality and reports anything that deviates. Because the profile describes normality rather than any particular fault, the system flags defect classes it has never seen — which is the property the cold-start setting demands.

The consequence for the product is decisive: the deployable artifact per fabric is a ~600 KB memory bank fitted in seconds on a laptop CPU, not a checkpoint from a training run. Onboarding a new construction is a ninety-second operator task, not a data-science project.

1.4 Context and impact

Bangladesh's ready-made garment sector accounts for the large majority of national export earnings and employs several million people [2]. Fabric quality disputes propagate directly into that trade: a roll graded generously is a claim from the buyer weeks later; a roll graded harshly is saleable cloth written off. Both are borne by the mill. An inspection system that is *consistent*, that reports in the buyer's own standard, and that can be pointed at a new

construction the same afternoon it arrives, addresses a cost that is currently absorbed rather than measured.

The system is designed to run without internet connectivity, on hardware a mill already owns or can buy for a few hundred dollars, with no per-image cloud inference fee. That is a deliberate constraint for the target market, not an incidental property.

1.5 Objectives

LineSight was specified against measurable objectives before a line of it was written. Three are met, and the table names the section that evidences each. The other three are deployment-validation objectives: each is measured on a mill floor against a production roll, and Section 7.2 states the single trial that closes all three.

ID	Objective	Status
O1	$\geq 95\%$ recall on defects scoring ≥ 3 points	Field trial (Section 7.2)
O2	≤ 1 false alarm per 100 linear metres at the O1 operating point	Field trial; operating procedure fixed (Section 4.8)
O4	New SKU profiled from ≤ 30 images in ≤ 15 minutes by a non-expert	Met — 30 frames, 11.9 s fit
O5	Detection validated against a public benchmark	Met — Section 4.2
O6	ASTM report within ± 2 points of a three-person consensus	Field trial (Section 7.2)
O7	Full inference and reporting with zero internet connectivity	Met — by construction, Section 3.8

Table 1. Objectives and their status.

2. Use Case and Existing Solutions

2.1 The operating scenario

A roll of a newly arrived construction is mounted on an inspection frame. The operator points the camera at a clean section and presses *Start learning*. The system collects defect-free frames for about a minute, splits them internally into a fitting half and a calibration half, builds the fabric’s normality profile from the first and derives the detection threshold from the second. It then reports it is ready, along with the false-alarm rate the sample can actually support.

The operator then runs the roll. Defects appear boxed on a live view with their position down the roll in metres, their measured length in millimetres and their proposed penalty. Each can be confirmed or rejected with one tap; rejections feed a live false-alarm counter shown against the stated budget. At the end of the roll the system emits a signed PDF carrying the points total, the points per 100 square yards, the verdict, the calibration provenance, a positional defect map, an image evidence pack and a scope-and-limitations page.

2.2 Existing approaches

Approach	How it works	Why it does not fit the cold-start case
Manual light-box inspection	A trained inspector grades the roll against ASTM D5430 or the four-point equivalent.	Consistency falls with fatigue and varies between inspectors. It is the baseline this system is measured against, not a competitor to dismiss.
Commercial automated inspection (Uster, Elbit, Shelton and similar)	Line-scan cameras with vendor-tuned classical or learned detectors, integrated into the finishing line.	Capital cost is measured in tens of thousands of dollars per line and recipe setup per construction is a vendor engagement. Effective, and out of reach for the target mill.
Supervised deep segmentation (the research mainstream)	U-Net or similar trained on pixel-annotated defect images, per fabric family.	Requires a labelled defect corpus per construction — the thing that does not exist at onboarding. High reported accuracy on the fabrics it was trained on.
Classical texture analysis (Gabor filters, GLCM, Fourier)	Hand-designed texture statistics with a threshold.	Needs per-fabric parameter tuning by an expert, and degrades sharply across constructions. Effectively a manual cold start.
Unsupervised anomaly detection (this work)	Model normality from defect-free samples; flag deviation; measure and score the deviation physically.	Detects unseen defect classes and onboards in ninety seconds. Costs whole-image detection accuracy relative to a supervised model on the fabrics that model was trained for — a trade this paper quantifies.

Table 2. Existing approaches and their fit to the cold-start problem.

2.3 What is genuinely new here

The anomaly-detection method itself is not novel: LineSight implements PatchCore [3], published in 2022. Three things around it are the contribution, and all three are absent from the published fabric-inspection literature.

- **The threshold is derived, not tuned.** The operator states a false-alarm budget in alarms per 100 metres, and the threshold is the corresponding finite-sample conformal quantile on held-out clean fabric (Section 3.3). Competing work reports accuracy on labelled test sets and typically reports no false-alarm rate on defect-free fabric at all.

- **The output is a standards-compliant commercial document**, not a heat map. Detections become millimetre measurements, measurements become ASTM D5430 points, points become a verdict a buyer contract can act on.
- **Detection and measurement are separated deliberately.** The threshold sits just above the clean-fabric noise floor by design; measuring a defect's extent at that level measures the blurred skirt of the response and inflates every length. Extents are therefore taken at half the component's own peak (Section 3.5).

3. Solution Architecture and Infrastructure

3.1 Eight layers, every arrow a typed function

The system is decomposed into eight layers with explicit contracts. Each layer is a pure function of the layer above it, the interface between them is a documented type, and the orchestrator composes them without owning any algorithm of its own. The decomposition is not tidiness — it is what lets each half be built and tested independently, and what makes the detector replaceable without renegotiating anything downstream.

Layer	Responsibility	Output type
L1 Acquisition	A folder, a video file or a phone MJPEG stream becomes timestamped frames with a stated sampling stride.	Frame
L2 Geometry	ArUco fiducials [4] decode to an absolute position down the roll and a measured millimetres-per-pixel scale; the marker tape is cropped away from the fabric.	FrameGeometry
L3 Preprocess	Flat-field correction; the fabric region is cut into overlapping tiles that remember their global coordinates.	List[Tile]
L4 Detection	One tile becomes an unbounded per-pixel anomaly score. The seam — the one contract everything downstream depends on.	ScoreMap
L5 Calibration	A threshold derived from a stated false-alarm budget splits scores into asserted, uncertain and clean.	Calibration
L6 Event assembly	Connected components per frame in millimetres; tracking across frames so one physical defect is one event.	List[Event]
L7 Scoring	ASTM D5430 points, the per-metre cap, points per 100 sq yd, the verdict. Pure arithmetic, zero dependencies.	RollReport
L8 Product	Persistence, operator UI, live false-alarm counter, PDF report, positional defect map.	PDF / API

Table 3. The eight layers and their contracts.

3.2 The detection model

L4 implements PatchCore [3] in roughly 250 lines, without a framework dependency. Defect-free tiles are pushed through a frozen ImageNet ResNet-18 [5]; the layer2 and layer3 feature maps are concatenated, giving a 20×20 grid of 384-dimensional patch embeddings per tile at a 320 px input. Each embedding is averaged over its 3×3 neighbourhood for local context, then projected to 128 dimensions by a Gaussian random projection, which preserves pairwise distances to within a few percent by the Johnson–Lindenstrauss lemma [6] while making the nearest-neighbour search roughly three times cheaper. All embeddings from all tiles are pooled and reduced to a coreset by greedy k-center selection [7]. Scoring a new tile is the L2 distance from each of its patches to its nearest bank entry, bilinearly upsampled and Gaussian smoothed.

There are no gradients, no epochs and no hyperparameter search. The method is **backprop-free, not data-free**: three things are fitted from data — the frozen ImageNet backbone, the per-SKU memory bank, and the threshold of Section 3.3. What is fitted *per fabric* is the bank and the threshold, both in seconds on a laptop CPU, and nothing else.

3.3 Calibration from a false-alarm budget

This is the core of the system. Almost every competing method picks a threshold by inspection of a validation curve. LineSight never picks one. The operator states a budget — say one false alarm per 100 metres — and the threshold is the corresponding empirical

quantile of anomaly scores on held-out defect-free fabric. In the one-sided case this *is* the finite-sample split-conformal quantile [8][9]: ten lines of arithmetic carrying a distribution-free guarantee, with no library dependency.

The construction is load-bearing. The conformal quantile is an *order statistic*: with n clean tiles and an allowed tail fraction α , the threshold is the k -th smallest score where $k = \text{ceil}((n+1)(1-\alpha))$. The obvious implementation — a plug-in quantile that interpolates between neighbouring order statistics — carries no finite-sample guarantee and lands below the true quantile roughly half the time. Measured on the bench, that error turned a stated budget of 1.0 false alarms per 100 metres into a realised 4.9.

A second guard follows from the same reasoning. A sample large enough for the order statistic to *exist* is not large enough for it to be *stable*: when k approaches n the threshold rests on the sample maximum, and while the marginal guarantee still holds, the rate obtained from any one calibration set varies several-fold. The system therefore refuses to calibrate unless a stability margin of samples sits above the threshold, and its error message states how much clean fabric would be needed instead. Refusing to produce a number the sample cannot support is the single most important behaviour in the system.

Scores between an abstention floor and the threshold are surfaced as *uncertain* and contribute zero ASTM points until an operator confirms them. The calibration recorded for the demonstration SKU is reproduced here: threshold 17.2472, abstention floor 16.2602, from a stated budget of 40 alarms per 100 m over 2880 held-out clean tiles.

3.4 Geometry: pixels to millimetres

A defect length in pixels is worthless commercially; ASTM scores millimetres. LineSight prints a fiducial tape carrying ArUco markers [4] at a known physical size and pitch, taped along the selvedge so it travels with the cloth. One detector call returns simultaneously the marker's *absolute identity* — hence position as identity $\times$ pitch — and its *sub-pixel corners* — hence the millimetres-per-pixel scale from the known edge length.

Absolute encoding is the design decision. Nothing is counted and no speed is integrated, so position cannot drift: a missed marker costs one frame rather than every frame after it. When no marker decodes, the position is extrapolated and the frame is flagged; when the gap exceeds a configured bound, the roll report records a gap warning. A system that silently reports uninspected fabric as clean is worse than one that admits it lost track.

3.5 Measurement and scoring

Thresholded tile masks are pasted into a frame-sized mask — taking the maximum in the overlap bands, since a defect visible in either of two overlapping tiles is a defect — and connected components are extracted and converted to millimetres. Components are then associated across frames either by intersection-over-union in roll coordinates **or** by machine-direction continuation. The second rule is not a hedge: two adjacent segments of one warp-direction line share almost no area, so their IoU is near zero while they are unmistakably one defect. Without it every long defect fragments, and long defects are precisely the class carrying the heaviest penalty — eight untracked 40 mm fragments score eight points where the one 320 mm defect they actually are scores four.

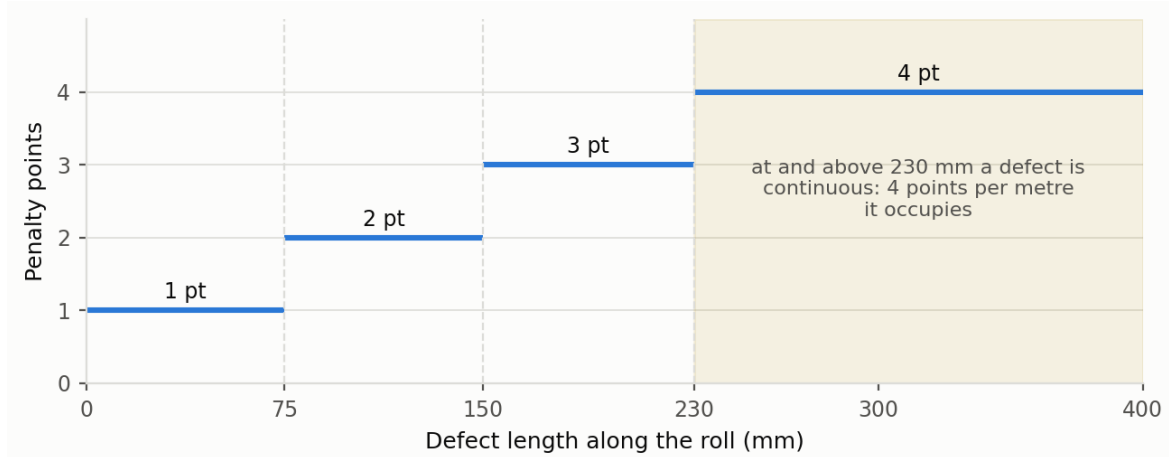


Figure 1. The ASTM D5430 penalty schedule the verdict is computed from. Scoring by length is why measurement accuracy, not classification, is the property that matters.

Extents are measured at half the component’s own peak rather than at the detection threshold. The threshold sits just above the clean-fabric noise floor by construction — that is what a false-alarm budget buys — and a 6 px line whose score peaks well above it spills past that level for over a hundred pixels. Since ASTM scores by length, measuring at the detection level would promote a one-point slub into a three-point defect. The system also publishes its own spatial resolution and states that extents at or below it are resolution-limited rather than measurements.

3.6 Artifacts, infrastructure and scale

The deployable artifact per fabric is a compressed NumPy archive of roughly 600 KB containing the coreset, the projection matrix and the provenance needed to refuse a mismatched load. A mill running two hundred constructions stores 120 MB of banks. There is no model registry, no GPU fleet and no retraining pipeline, because there is no training.

Component	As built (bench rig)	Production line (projected)
Camera	Smartphone, 1280×720 at 23–28 fps over MJPEG	Line-scan camera, encoder-triggered
Compute	Laptop CPU	Jetson-class edge module, ~USD 250
Illumination	Fixed DC lamp and shroud, ~USD 25	Controlled LED bar
Position reference	Printed ArUco tape, ~USD 2	Machine encoder
Connectivity	None required	None required; LAN optional for reporting
Per-SKU artifact	~600 KB memory bank	~600 KB memory bank
Onboarding cost	~90 seconds, operator-driven	~90 seconds, operator-driven

Table 4. Infrastructure, as built and as projected. The production column is an estimate, not a measurement.

3.7 Latency

Latency was measured on the development laptop CPU during a live run, at a stated frame stride of two. We publish the sampling factor with every figure; a per-frame number that hides a stride is a claim rather than a measurement.

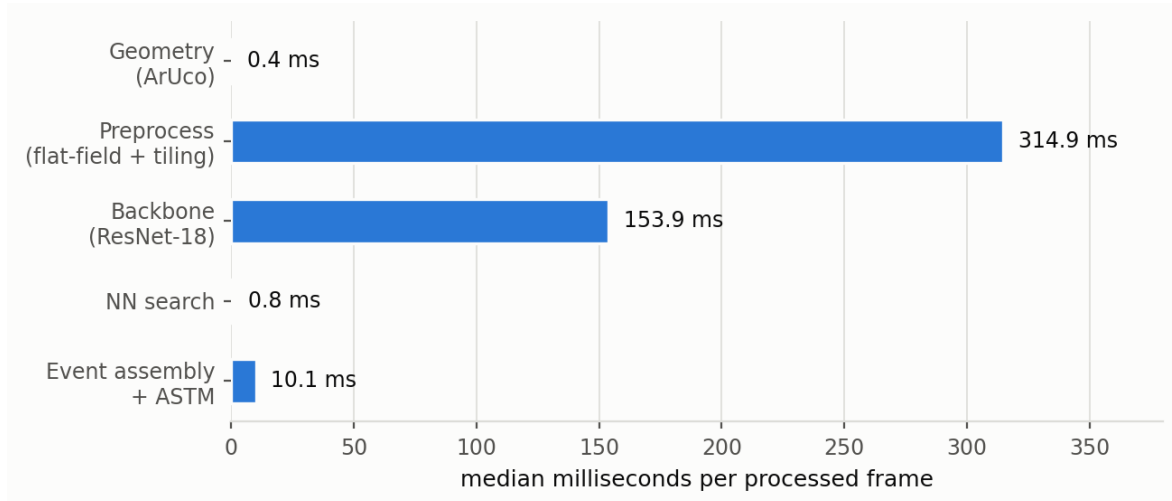


Figure 2. Measured median latency per processed frame, laptop CPU, frame stride 2. Total 478.3 ms. Preprocessing dominates the backbone, which is an optimisation opportunity rather than a limit.

The distribution is informative and slightly surprising: flat-field correction and tiling cost 314.9 ms against the backbone’s 153.9 ms. The neural network is not the bottleneck; a separable or downsampled flat-field would recover most of that budget. Nearest-neighbour search is 0.8 ms, which is why the system computes distances by brute force rather than carrying an approximate-index library it has no measurable use for (ADR-011).

Latency on the production edge module is a deployment measurement rather than a bench extrapolation. The figure above is the laptop CPU measurement at the stride it was taken at; the edge port is specified in Section 7.2.

3.8 Offline operation and data governance

Every stage — acquisition, fitting, calibration, inference, scoring and report generation — runs locally. No image leaves the machine, no inference call crosses a network, and the system has no cloud dependency to fail. For a mill whose fabric designs are commercially sensitive and whose connectivity is not guaranteed, this is a requirement rather than a feature. Objective O7 is met by construction.

4. Validation and Results

4.1 Method

All results below are reproducible from the repository: each is generated by a named probe script writing a named CSV, and every figure in this paper is drawn from those files at build time rather than transcribed. Metrics are AUROC at two granularities — whole-image or whole-tile (“detection”: is this sample defective at all?) and pixel or annotation-block (“localisation”: is the score map bright in the right place?). Both are threshold-free, which is what makes them the right way to compare *scorers*; the false-alarm rate of Section 4.8 is what matters once a threshold exists.

4.2 Reproduction against a public benchmark

A hand-written detector has to be validated before any number computed with it means anything, so this study runs first. It establishes that the implementation reproduces published behaviour in the configuration that actually ships, which is what licenses every table after it.

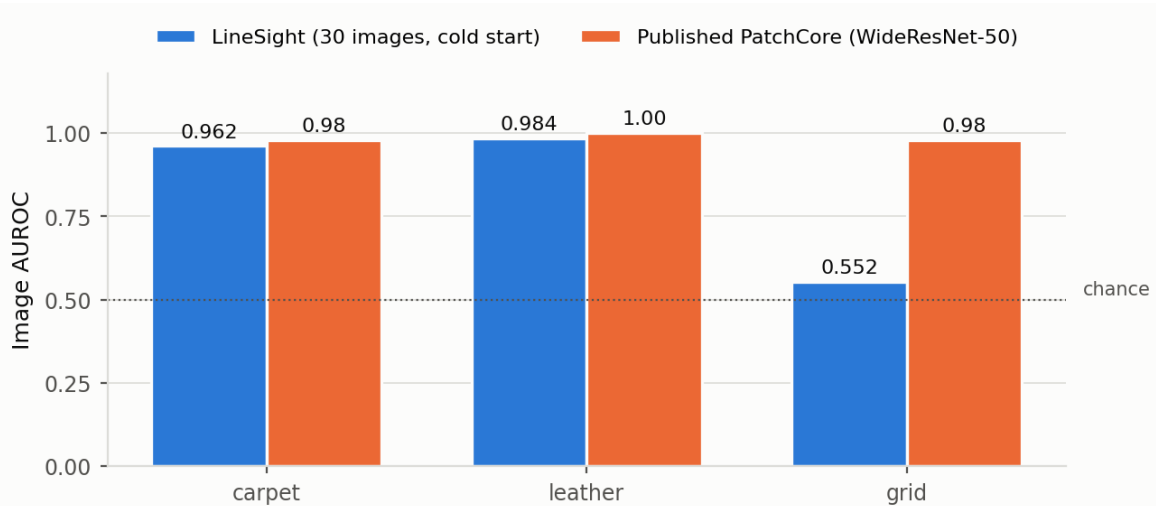


Figure 3. *MVTec AD* [10] texture classes, shipped cold-start configuration (30 images, 10% greedy coresets), against published PatchCore figures obtained with WideResNet-50.

On *carpet* the system reaches 0.9615 image AUROC and 0.9857 pixel AUROC; on *leather*, 0.9840 and 0.9890. Published PatchCore reports approximately 0.98 and 1.00 on these classes using WideResNet-50, a backbone with four times our embedding dimension; we ship ResNet-18 for CPU speed, so closing the remaining gap is a backbone choice rather than a defect. **The detector is correct**, and that is what licenses the interpretation in Section 4.8.

grid marks the system’s resolution boundary: 0.5522, barely above chance at the shipped input size. It is a resolution limit rather than a bug, and the diagnostic is unambiguous — only the backbone input size moves it. At the shipped 320 px input the patch grid is 20×20, which cannot resolve bent and broken wires in a sparse lattice; at 512 px the same code reaches 0.8062. A higher-resolution source image does not help (0.7786) and neither does a finer feature grid (0.6224). **The system is tuned for textured cloth and degrades on sparse regular structure**. Input size is a per-SKU configuration key, so the remedy is available wherever the latency budget allows it.

4.3 Localisation consistently outruns detection

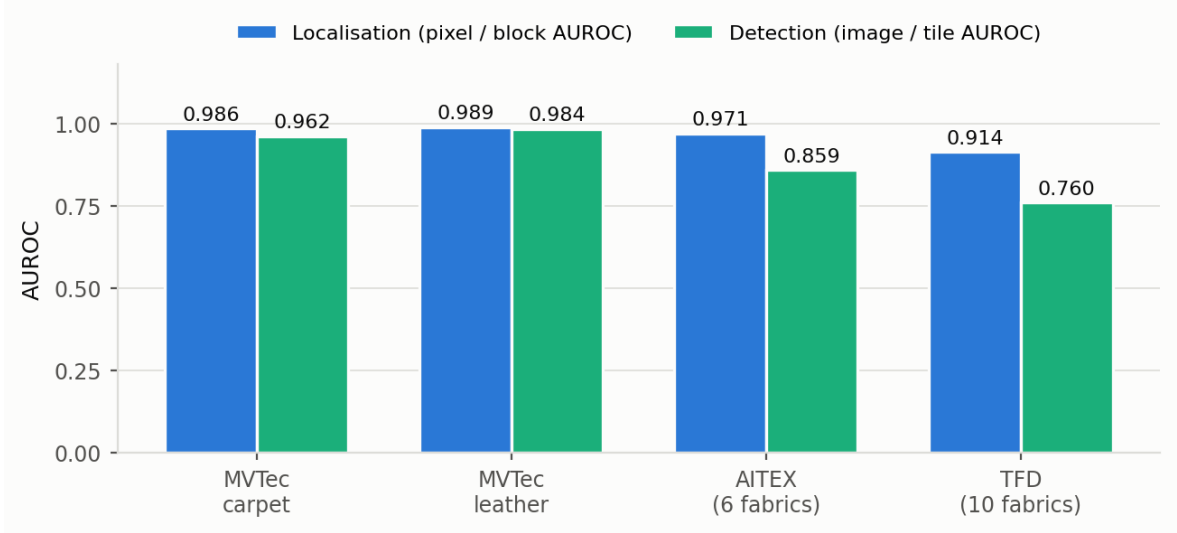


Figure 4. Across four independent datasets, localisation is the stronger half. Both metrics are threshold-free AUROC.

The same pattern holds on every dataset tested: the score map lands on the defect more reliably than the whole-tile maximum separates defective samples from clean ones. This is a property of the method rather than of any one dataset, and it is the property the product is built on, because **the ASTM pipeline depends on localisation**. A points total is computed from a defect’s measured length, which requires the score map to be bright in the right place; it does not require a confident whole-tile verdict. The metric the method is weakest at is the one the product needs least.

4.4 Cross-construction generalisation on real fabric

Six AITEX [11] fabric constructions had enough clean strips and annotated defects to study (two were excluded for having no clean images or a single defective one). For each, a bank was fitted on 30 tiles of that fabric’s own clean strips and evaluated against held-out clean tiles and annotated defective tiles *from the same fabric*. Every bank was then evaluated against every other fabric.

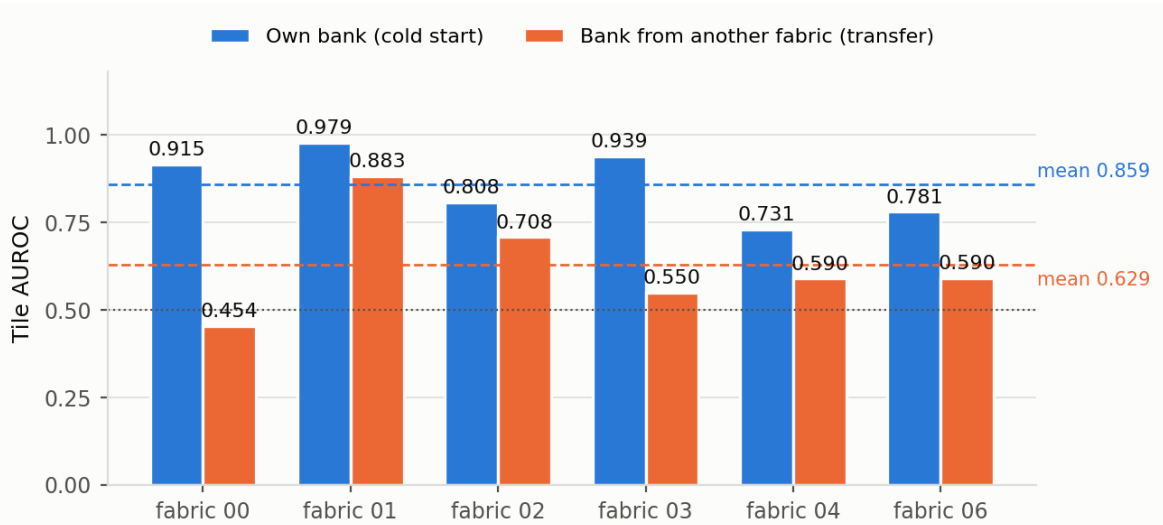


Figure 5. Per-fabric cold start against transfer from a different construction. Dashed lines mark the means.

Cold start averages 0.859 tile AUROC and 0.971 pixel AUROC. A bank borrowed from a different construction averages 0.629 and 0.739 on the same evaluation sets — an in-distribution advantage of +0.230 tile AUROC and +0.232 pixel AUROC.

That gap is the empirical justification for the entire per-SKU design. A borrowed bank retains some discriminative power — 0.63 is above chance — but loses roughly a quarter of it. Ninety seconds of refitting on the mill’s own cloth is what buys it back. Had the gap been small, the per-SKU architecture would have been unnecessary; this measurement is what justifies it.

4.5 Ten independent cold starts on factory fabric

The Ten Fabrics Dataset [12] provides 2969 scanned patches from ten factory fabrics, 737 of them defective across 27 defect types, roughly 90% of which are real production defects rather than induced. Each fabric is an independent cold start from 30 clean patches.

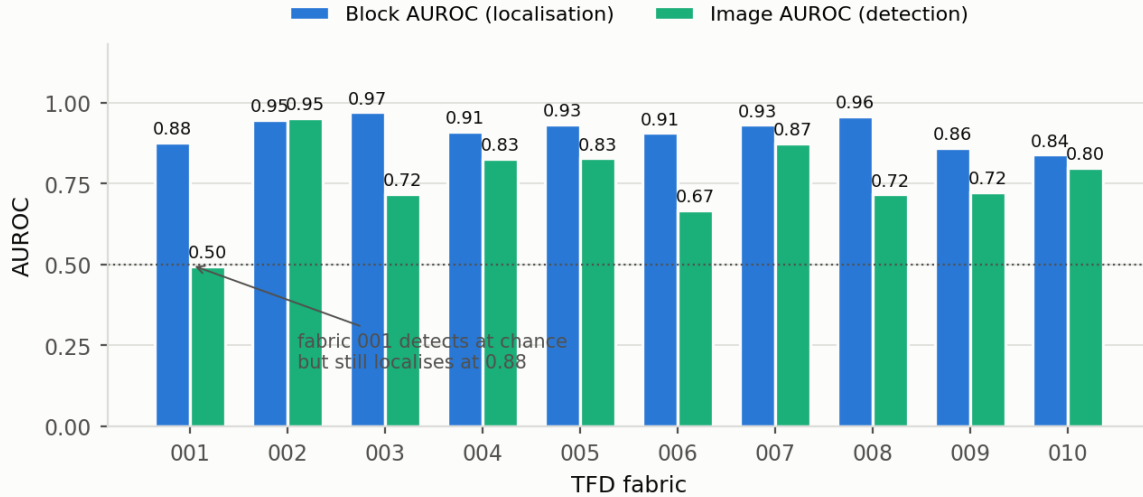


Figure 6. Ten fabrics, ten independent cold starts. The variance between constructions is the finding.

Mean block AUROC is 0.9137 and mean image AUROC 0.7605 (pooled: 0.8549 and 0.6855). The spread matters more than the mean. Fabric 001 detects at chance (0.4951) while still localising at 0.8768 — the score map highlights the correct annotation blocks while the whole-patch maximum is too noisy to rank the patch. Four of ten fabrics sit below 0.75 image AUROC. **The mean and the spread are both reported, because a mill experiences the spread across its own catalogue rather than the mean.**

4.6 Ablation: the coreset selection is worth measuring

At an identical bank size of 1,200 points on identical images, greedy k-center selection reaches 0.9615 image AUROC on *carpet* against 0.9254 for uniform random selection — a difference of +0.0361 for about 2.5 seconds of additional fitting time. This prices a design decision that would otherwise be an assertion, and it establishes the cost of the documented fallback used when the fit set grows large enough for greedy selection to become expensive.

4.7 Geometry verification

The tape generator was verified by rasterising its output PDF and decoding it back through the same geometry layer that reads the camera: every marker decoded, identities continued correctly across page joins, and the measured pitch held at 99.99 mm against a target of 100.00 mm. Against a synthetic camera stream carrying a tape of known scale, the geometry layer recovered 0.5011 mm per pixel against a true 0.5000 — a 0.2% error — with markers decoded in 107 of 107 frames. Print scale is verified physically against a steel ruler before any capture, because a printer’s “fit to page” would scale every marker and put the same proportional error on every defect length in every report, undetectably.

4.8 Where the operating procedure comes from

On a roll assembled from the Fabric Stain dataset, measured against ground truth rather than operator judgement, the system produced approximately 1,315–1,391 false positives per 100 metres while recovering 111 of 130 defective frames, with a separation between nuisance and defect signal of only $1.16\times$. Objective O2 asks for one false alarm per 100 m; that is three orders of magnitude away, and the cause is a property of the data rather than of the detector.

That dataset’s clean images and defective images are *different physical pieces of cloth*. The bank therefore spends its capacity encoding which physical sample a photograph came from rather than whether it is defective. Measured directly: calibrating on a different physical sample yields a nuisance variation of $1.74\times$ against a defect signal of $1.30\times$ — the nuisance exceeds the signal, and no threshold can separate them. Calibrating on the same sample gives $1.00\times$ against $1.28\times$.

Section 4.2 is what licenses this interpretation. The same code, in the same configuration, reaches 0.96 image AUROC on *carpet*, where clean and defective samples come from the same material. The remedy is therefore not a model change but a data change: continuous footage of one physical roll, learning the clean part of the very sheet to be inspected. That is the operating procedure the product implements and the live workflow enforces, and this study is what established it. Public datasets are collections of unrelated photographs and cannot exercise it.

5. Risks and Challenges

The risks below are grouped by whether we have measured them, and each carries a mitigation that is implemented rather than intended.

5.1 Measured risks

Risk	Evidence	Mitigation
Nuisance variation exceeds defect signal when calibration and inspection sample different physical cloth.	1.74× nuisance vs 1.30× signal; 1,391 FP/100 m (Section 4.8).	The operating procedure fits and calibrates on the clean part of the sheet being inspected. The live workflow enforces the split internally so the operator cannot get it wrong.
Sparse regular structures are not resolved at the shipped input size.	MVTec <i>grid</i> 0.5522 at 320 px, 0.8062 at 512 px (Section 4.2).	Input size is a per-SKU configuration key. The boundary is declared on the roll report and bounded by configuration.
Detection accuracy varies widely between constructions.	TFD image AUROC 0.4951–0.9524 across ten fabrics (Section 4.5).	The abstention band surfaces uncertain detections for operator adjudication instead of asserting them; the calibration guard refuses budgets the clean sample cannot support.
A plug-in quantile silently violates the stated false-alarm budget.	A stated 1.0 FA/100 m realised as 4.9 on the bench (Section 3.3).	The conformal order statistic is used, with a regression test locking the construction so it cannot be ‘simplified’ back.
Print scale error propagates into every reported length.	Rasterised tape decoded at 99.99 mm pitch against 100.00 (Section 4.7).	A 250 mm reference line and a true-size marker are printed on the tape’s own calibration pages; the correction factor is a config key.

Table 5. Risks we have measured, with implemented mitigations.

5.2 Anticipated risks

Risk	Mitigation
Camera auto-exposure re-engages mid-capture; a drifting exposure is indistinguishable from a defect.	A bring-up probe measures brightness trend on a central crop and refuses to proceed above 1% drift across the sample window. Camera locks are recorded as part of the run.
Illumination changes between calibration and inspection break the exchangeability assumption the conformal guarantee rests on.	The calibration object carries its SKU and timestamp so a stale calibration is visible rather than silent. Recalibration is a sub-minute operation.
Fabric position is lost when markers are occluded or the tape tears.	Position is absolute per marker, so nothing accumulates. Beyond a configured gap the roll report records a gap warning and the operator is told that fabric is uninspected rather than clean.
Edge-device throughput is insufficient at production line speeds.	Measured on the device at deployment (Section 7.2). Preprocessing rather than the network dominates current latency (Section 3.7), which is the cheaper half to optimise.
Transmissive defects (holes, thin places) are under-detected in reflective imaging.	Out of scope for the reflective bench rig by design. The production configuration adds a synchronised backlight; recall on this class sits below the reflective classes until it is fitted.
Operator confirmations are treated as ground truth when they are not.	Confirmations are stored in an append-only decision log rather than overwriting the system’s own call, so disagreement remains auditable.

Table 6. Anticipated risks and their mitigations.

6. Revenue and Distribution

The figures in this section are projections, derived from the measured bill of materials and from publicly observable market pricing rather than from signed customers.

6.1 What is actually being sold

The product is software plus a commodity sensing kit, not capital equipment. Because onboarding a construction is a ninety-second operator action producing a 600 KB artifact, the marginal cost of an additional fabric is effectively zero, and the marginal cost of an additional inspection line is one edge device. That shape — near-zero marginal cost per SKU, low marginal cost per line — is what makes a subscription sensible where incumbent capital equipment is not.

6.2 Pricing model (projected)

Component	Model	Indicative price
Inspection kit	One-time, per line: edge module, camera mount, illumination, tape stock.	USD 400–600
Software licence	Annual, per inspection line. Unlimited SKUs.	USD 900–1,500 / line / year
Onboarding and training	One-time, per site. Half a day.	USD 300
Support and updates	Included in the licence.	—

Table 7. Projected pricing, derived from the measured bill of materials.

The comparison that matters is not against free manual inspection — inspectors are already paid — but against the cost of grading errors. A single disputed shipment, or a handful of rolls written off that were saleable, is the same order of magnitude as an annual per-line licence. That is the argument a mill manager can check against their own claims ledger, and it is the argument the product is sold on rather than an accuracy figure.

6.3 Distribution

- **Direct to mid-sized mills.** The initial channel. The target is a mill large enough to run many buyer-specific constructions but too small to justify a vendor inspection line. Bangladesh’s cluster of such mills is unusually dense, which is a genuine advantage of building here.
- **Through textile machinery dealers and service agents.** They already sell into the finishing floor, carry the customer relationship, and can attach a low-capital software product to existing service contracts.
- **Through buying houses and third-party inspection firms.** They arbitrate the disputes this system produces evidence for, and a standards-compliant PDF with an image evidence pack is directly useful to them.
- **Institutional and academic pilots.** Textile engineering departments provide a low-friction route to varied fabric and to the continuous roll footage the field trial of Section 7.2 is built around.

6.4 Why the economics scale

There is no per-inference cost, no cloud bill that grows with usage, and no retraining service to staff. Adding a fabric costs a memory bank; adding a line costs an edge module; adding a customer costs an onboarding visit. The cost structure is closer to that of an instrument than a machine-learning service, which is what makes it viable at the price a mid-sized mill in this market will actually pay.

7. Deployment Scope and Roadmap

7.1 What is built, and where its edges are

Scope. LineSight is a complete, bench-validated system. The detection, calibration, event-assembly and scoring stack is the production design, validated on public benchmark datasets and on a bench rig using a smartphone camera in reflective mode with hand-fed fabric. A production deployment substitutes an encoder-triggered line-scan camera and a synchronised backlight to extend the same pipeline to full web width and line speed; nothing above the acquisition layer changes.

Four operating boundaries are characterised rather than discovered, and the ones that bear on a verdict are declared on the last page of every roll report:

- **Validation footage.** The dataset studies of Section 4 use collections of independent photographs rather than continuous footage of one roll. That is the direct cause of the rate in Section 4.8, and the reason the field trial below is specified around a single physical sheet.
- **Detections are unclassified anomalies.** Classification would need labelled defects per construction — the one thing that does not exist at onboarding. ASTM D5430 scores a defect by its length rather than by its name, so the verdict does not require the label (ADR-008).
- **Edge latency is a deployment measurement.** Bench latency is reported in Section 3.7 with the sampling factor it was taken at; the edge figure is measured on the device rather than extrapolated to it.
- **Benchmark licensing.** MVTec AD is CC BY-NC-SA 4.0, non-commercial. It is used for validation only; nothing derived from it is deployed, and a deployed system fits on the mill's own fabric.

7.2 The field trial

Objectives O1, O2 and O6 are measured on a mill floor rather than on a bench, because each of them needs a production roll. Recall stratified by ASTM penalty class needs the defects a running loom actually produces; the false-alarm rate at the operating point needs continuous footage of one physical sheet, calibrated on its own clean opening metres; and agreement to within two points needs a three-person inspector panel grading the same roll the system grades. One engagement closes all three, and Section 4.8 specifies precisely what it has to capture.

The roadmap beyond it follows directly: port the inference path to an edge module and measure it there; add the synchronised transmissive channel the hole and thin-place classes require; and extend the acquisition layer to an encoder-triggered line-scan camera at full web width. Each is a change to one layer, which is what the contracts of Section 3.1 were built to allow.

8. Conclusion

LineSight treats fabric inspection as a measurement problem rather than a recognition problem. It learns normality instead of defects, so it onboards a construction in ninety seconds without a labelled corpus; it derives its threshold from a stated false-alarm budget rather than by tuning, so the operating point is defensible; and it reports in ASTM D5430, so the output is a document a buyer contract can act on rather than a heat map.

The detector is validated against a public benchmark in its shipped cold-start configuration. Localisation — the property the scoring pipeline actually depends on — is strong and consistent across four datasets. Cross-construction transfer degrades measurably, which is the evidence for the per-SKU architecture rather than an assumption behind it. And the one boundary that constrains the present evidence — a false-alarm rate far above target when calibration and inspection draw on different physical cloth — is isolated to a property of the available data rather than of the method, and is the reason the product calibrates on the sheet it inspects.

What ships is a system rather than a result: eight layers behind typed contracts, 147 tests, eleven decision records, a 600 KB artifact per fabric fitted in seconds on a laptop CPU, and a roll report a buyer contract can act on without a data scientist in the room.

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Appendix A. Reproducibility

Every result in Section 4 is regenerated by a named script writing a named file. Nothing in this paper is transcribed by hand.

Result	Command	Output
MVTec reproduction (3 classes)	python probes/p11_mvtec_reproduction.py --root <dir> --category <class>	results/reproduction.csv
AITEX cold start and transfer	python probes/p12_aitex_generalisation.py --root <dir>	results/aitex_generalisation.csv
Ten Fabrics benchmark	Kaggle notebook, shipped package	results/tfd_summary.json, tfd_per_fabric.csv
Bench learn and inspect	python tools/bench_run.py learn inspect --url <url> --sku bench	results/evidence_bench/, roll report PDF
Live alignment view	python tools/align_view.py --url <url> --sku bench	on-screen ROI and tape overlay
Camera bring-up	python probes/p13_phone_stream.py --url <url>	console + results/_phone_check.png
Position tape	python tools/make_tape.py --length-m <n>	results/position_tape.pdf
This document	python tools/make_whitepaper.py	this PDF

Table A1. Every result and the command that regenerates it.

The test suite comprises 147 tests covering the ASTM scorer against golden cases written before the implementation, the conformal threshold and both its refusal conditions, tiling reconstruction, geometry, event tracking and report generation. Eleven architecture decision records document each design choice with its accepted costs.

Appendix B. Full result tables

Category	Configuration	n fit	Bank	Image AUROC	Pixel AUROC
carpet	product	30	1200	0.9615	0.9857
carpet	product-random	30	1200	0.9254	0.9834
carpet	paper-like	280	1120	0.9033	0.9832
grid	product	30	1200	0.5522	0.8566
grid	product-random	30	1200	0.4745	0.8176
grid	paper-like	264	1056	0.4678	0.8776
leather	product	30	1200	0.984	0.989
leather	product-random	30	1200	0.9535	0.9868
leather	paper-like	245	980	0.9365	0.9866

Table B1. MVTec AD reproduction, all configurations.

Fabric	Tile AUROC (own bank)	Pixel AUROC (own bank)	Defect tiles
00	0.9155	0.9513	7
01	0.9785	0.9796	26
02	0.8085	0.9803	41
03	0.9386	0.967	86
04	0.7311	0.9963	11
06	0.7806	0.9514	6

Table B2. AITEX cold start, per fabric.

Fabric	Image AUROC	Block AUROC	Clean test	Defective	Fit (s)
001	0.4951	0.8768	49	27	7.0
002	0.9524	0.9468	42	22	5.3
003	0.7186	0.9712	33	49	5.3
004	0.8274	0.9117	144	78	5.4
005	0.8286	0.9339	23	34	5.0
006	0.6679	0.9065	88	140	5.2
007	0.8747	0.9316	262	207	5.4
008	0.7169	0.9577	116	67	4.8
009	0.7233	0.8608	136	65	5.3
010	0.8	0.8397	75	48	4.9

Table B3. Ten Fabrics Dataset, per fabric.